Structure-Aware Coarse-to-Fine Graph Reconstruction

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***Abstract*—Large-graph generation remains challenging because directly modelling all possible node pairs is computationally expensive, while coarse representations often discard structural information needed for accurate reconstruction. We propose a structure-aware coarse-to-fine graph synthesis framework that decomposes graph generation into a collection of smaller local reconstruction problems. We first use K-MIS coarsening to partition the graph into connected, localised supernodes and construct a compact graph synopsis. Beyond supernode sizes and edge counts, the synopsis retains a centroid-relative hop distribution h_i to describe intra-supernode structure and cross-degree distributions $r_{i\to j}$ to characterise how edges are distributed between supernodes. We analyse the structural properties of the K-MIS partition and show how these summaries reduce the space of feasible fine-grained reconstructions. Based on this representation, we develop two conditional discrete diffusion models: D_{intra} reconstructs the internal topology of individual supernodes, while D_{inter} reconstructs edges between connected supernodes. The resulting blocks are then assembled into a complete graph. Our formulation supports both source-to-target transfer using only a target coarse synopsis and adaptation using limited target fine-grained structure. Experiments on road, collaboration, and citation networks evaluate local reconstruction fidelity, structural conditioning, cross-graph transfer, end-to-end graph quality, and scalability. [Final results sentence to be completed once all experiments finish.]**

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

- Large graph generation is difficult because full graph diffusion is expensive.
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- we propose a two-stage conditional graph diffusion framework that reconstructs the fine graph from the coarse synopsis.
- We formulate coarse-to-fine graph synthesis as a stochastic graph uncoarsening problem and propose a two-stage conditional diffusion framework for intra- and inter-supernode reconstruction.
- We introduce permutation-invariant structural summaries for recovering information lost by coarsening: a radial hop distribution h_i for intra-supernode structure and a cross-degree distribution $r_{i\to j}$ for inter-supernode structure.
- We evaluate structural fidelity, diversity, cross-graph transfer, and scalability across road, social, and collaboration networks.
- HiGGs itself even notes that a partitioning method producing more near-uniform partitions could reduce the maximum training-graph size and allow smaller models.

Huiping: 5. avoid self-loop for $\mathrm{D}_{intra} \to jamie$

II. Related Work

Graph generation and graph diffusion. Deep graph generators model distributions over discrete, permutation-invariant graph structures. Early autoregressive methods generate graphs sequentially, including GraphRNN [28] and GRAN [16]. Later score-based and diffusion models avoid a fixed generation order. Niu et al. [22] introduced permutation-invariant score-based graph generation, followed by GDSS [13], GraphGDP [11], and the discrete diffusion model DiGress [27]. Structural guidance has also been explored: SPECTRE [19] uses Laplacian spectral information, while EDGE [4] exploits sparsity and degree guidance for larger graphs. These studies show that structural information can improve graph generation and constrain the generation space.

Hierarchical / scalable graph generation. Hierarchical generation reduces the cost of modelling all $O(n^2)$ potential edges. BiGG exploits sparsity through a hierarchical autoregressive representation, while hierarchical molecular generators construct graphs from larger structural motifs [5], [12]. HiGen [14] generates graph communities and their cross-community edges separately, while HiGGs [6] provides a model-agnostic hierarchical framework and demonstrates generation of graphs with tens of thousands of nodes using DiGress. HSpectre [2] uses repeated local diffusion-based expansion for subquadratic generation. More recent methods perform generation directly over coarse representations: SBGD [25] uses stochastic blocks, LGDC [23] uses spectrum-preserving graph coarsening followed by expansion, and FAHNES [9] combines repeated coarsening with localised graph and hypergraph expansion.

Graph coarsening and summarisation. Graph summarisation represents large graphs using compact structures. SNAP and k-SNAP provide controllable graph aggregation [26],

while Navlakha et al. use supernodes, superedges, and correction sets for lossless or bounded-error reconstruction [20]. SWeG extends supernode-based summarisation to web-scale graphs [24]. Spectral coarsening instead aims to preserve spectral and cut properties, with theoretical guarantees developed by Loukas and Vandergheynst and Loukas [17], [18].

III. Problem Formulation and Overview

A. Graph Notation and K-MIS Coarsening

Let $G = (V, E)$ be an undirected, unweighted, simple graph, where V and E denote the vertex and edge sets, respectively. Let $n = |V|$ and $m = |E|$, and let $\mathbf{A}$ denote the adjacency matrix of G. We adopt K-MIS graph coarsening, where K is a locality parameter that controls the coarsening resolution. Given G, K-MIS selects a set of centroids $S = \{s_1, \dots, s_M\}$ such that the shortest-path distance between any two centroids is greater than K, while every vertex is within K hops of at least one centroid.

Each vertex is assigned to its nearest centroid, with ties resolved using a fixed ordering. Let $\pi : V \to S$ denote this assignment. The vertices assigned to centroid s_i form a supernode $V_i = \{v \in V : \pi(v) = s_i\}$. The resulting supernodes $\{V_1, \dots, V_M\}$ form a partition of V.

Let $G_i = G[V_i]$ denote the subgraph induced by V_i, with $n_i = |V_i|$ vertices and m_i internal edges. We represent the partition using a binary assignment matrix $\mathbf{P} \in \{0,1\}^{M \times n}$, where $\mathbf{P}_{iv} = 1$ if vertex v belongs to supernode V_i, and 0 otherwise. The weighted coarse adjacency matrix is defined as

$$\mathbf{A}_c = \mathbf{P}\mathbf{A}\mathbf{P}^\top.$$

Its diagonal entry $(\mathbf{A}_c)_{ii} = 2m_i$ gives twice the number of internal edges in supernode V_i, while an off-diagonal entry

$$w_{ij} = (\mathbf{A}_c)_{ij}$$

gives the number of edges between supernodes V_i and V_j.

The resulting weighted coarse graph is denoted by $G_c = (S, E_c, \mathbf{A}_c)$, where E_c contains pairs of supernodes with non-zero cross-edge counts. We further denote the internal adjacency matrix of G_i by $\mathbf{L}_i$ and the bipartite adjacency matrix between supernodes V_i and V_j by $\mathbf{B}_{ij}$.

B. Structure-Aware Coarse Representation

The coarse graph G_c preserves supernode connectivity and edge counts, but does not specify how edges are arranged within or between supernodes. To retain additional structural information for fine-grained reconstruction, we introduce three structural summaries.

Intra-supernode density. For each supernode V_i, we define its internal edge density as

$$\rho_i = \begin{cases} \dfrac{2m_i}{n_i(n_i-1)}, & n_i > 1, \\ 0, & n_i = 1. \end{cases}$$

The pair (n_i, ρ_i) specifies the size and density of the induced subgraph G_i, but does not determine its internal topology.

Hop-distance distribution. To characterise the radial structure of a supernode, we define $\mathbf{h}_i = (h_i(1), \dots, h_i(K))$, where

$$h_i(\ell) = \frac{|\{v \in V_i \setminus \{s_i\} : d_G(v, s_i) = \ell\}|}{n_i - 1}, \qquad \ell = 1, \dots, K,$$

for $n_i > 1$, and $\mathbf{h}_i = \mathbf{0}$ for $n_i = 1$. Here, $h_i(\ell)$ gives the proportion of non-centroid vertices at hop distance ℓ from centroid s_i. This summary captures how vertices are distributed around the centroid and therefore provides structural information beyond supernode size and density.

Cross-degree distribution. For two connected supernodes V_i and V_j, we define the cross-degree of a vertex $u \in V_i$ towards V_j as

$$d_{i \to j}(u) = |\{v \in V_j : (u, v) \in E\}|.$$

We denote by $\mathbf{r}_{i \to j}$ the normalised histogram of these cross-degrees over vertices in V_i. We use five degree bins, with bin boundaries adapted to the degree scale of the graph based on its median degree. This allows the summary to be applied to graphs with different degree distributions while keeping a fixed-dimensional representation. While w_{ij} records only the total number of edges between V_i and V_j, $\mathbf{r}_{i \to j}$ captures how these edges are distributed across the vertices in V_i.

We refer to the weighted coarse graph together with the supernode sizes, densities, hop-distance distributions, and cross-degree distributions as the *structure-aware graph synopsis*, denoted by $\mathcal{S}_K(G)$. The basic statistics (n_i, ρ_i, w_{ij}) capture graph size and edge quantities, while $\mathbf{h}_i$ and $\mathbf{r}_{i \to j}$ provide additional information about the arrangement of intra- and inter-supernode edges. These summaries are subsequently used as structural conditions for fine-grained graph reconstruction.

C. Problem Formulation

We study *coarse-to-fine graph synthesis*, where the goal is to generate a fine-grained graph from its structure-aware coarse representation. Let G_s denote a source graph available for training and G_t an unseen target graph. We assume that the fine-grained source graph G_s is available during training, while only the structure-aware synopsis $\mathcal{S}_K(G_t)$ is required to synthesise the target graph.

During training, K-MIS coarsening is applied to G_s to construct two types of training instances: intra-supernode subgraphs G_i and inter-supernode adjacency blocks $\mathbf{B}_{ij}$, together with their corresponding structural conditions. We use two conditional discrete diffusion models. The intra-supernode model, D_{intra}, learns to generate the internal adjacency structure of each supernode conditioned on its size n_i, density ρ_i, hop-distance distribution $\mathbf{h}_i$, and coarsening parameter K. The inter-supernode model, D_{inter}, learns to generate edges between connected supernodes conditioned on their structural statistics, cross-edge count w_{ij}, and cross-degree distributions $\mathbf{r}_{i \to j}$ and $\mathbf{r}_{j \to i}$.

At inference time, D_{intra} first generates an internal subgraph $\widehat{G}_i$ for each target supernode using the conditions provided by $\mathcal{S}_K(G_t)$. Given the generated intra-supernode structures,

D_{inter} then generates the corresponding inter-supernode adjacency blocks $\widehat{\mathbf{B}}_{ij}$. The generated intra- and inter-supernode blocks are finally assembled to form the complete synthetic target graph $\widehat{G}_t$.

The models therefore learn from G_s how to map coarse structural information to fine-grained graph structures and apply this learned mapping to $\mathcal{S}_K(G_t)$. Since multiple fine-grained graphs can share the same coarse synopsis, recovering the exact edge arrangement of G_t is generally underdetermined. Our goal is therefore not exact graph recovery, but to generate graphs that satisfy the supplied coarse statistics while preserving important local and global structural properties of the target graph.

Our central question is how the additional structural information provided by $\mathbf{h}_i$ and $\mathbf{r}_{i\to j}$ reduces reconstruction ambiguity and improves the structural fidelity of the generated graphs.

IV. Coarse Graph Representation

- Prove what K-MIS guarantees about each supernode(check the downsampling paper)
- add a bound on supernode size

Our framework uses K-MIS [1] coarsening to decompose a large graph into a collection of local reconstruction regions. While the coarse graph preserves the relationships between these regions, the standard coarse representation retains limited information about the fine-grained topology within and across supernodes. We therefore augment the K-MIS representation with structural summaries that provide additional guidance for coarse-to-fine reconstruction. In this section, we describe the resulting graph representation and analyse its structural properties and reconstruction ambiguity.

A. K-MIS-*Based Graph Partition*

Building on the K-MIS formulation introduced in Section III, we construct the centroid set $S = \{s_1, \ldots, s_M\}$ using a degree-ranked greedy procedure. Vertices are ordered according to their degree, and the highest-ranked eligible vertex is selected as a centroid until a maximal K-independent set is obtained. The degree-based ordering determines which valid K-MIS is selected, while the separation and coverage properties of K-MIS remain unchanged. After centroid selection, each vertex is assigned to its nearest centroid, with ties resolved using the same fixed centroid ordering. This produces a partition

$$V = V_1 \cup \cdots \cup V_M, \qquad V_i \cap V_j = \emptyset \quad \text{for } i \neq j,$$

where V_i denotes the vertices assigned to centroid s_i. Each induced subgraph $G_i = G[V_i]$ forms a supernode and serves as a local reconstruction unit for D_{intra}. The resulting weighted coarse graph G_c contains one node for each supernode. Its diagonal entries record the number of internal edges m_i, while an off-diagonal weight w_{ij} records the number of original edges connecting V_i and V_j. Hence, the K-MIS partition converts the original graph into a set of local subgraphs together with a coarse connectivity structure specifying which supernode pairs must subsequently be reconstructed by D_{inter}.

B. Structural Properties and Size Reduction

The K-MIS partition provides several useful structural properties for coarse-to-fine reconstruction. We first state the locality property of the resulting supernodes.

Proposition 1 (Locality of K-MIS supernodes)**.** *Let V_i be the nearest-centroid partition induced by a maximal K-independent set, with a fixed tie-breaking order. For every $v \in V_i$,*

$$d_G(v, s_i) \leq K.$$

Moreover, there exists a shortest path from v to s_i containing only vertices in V_i. Consequently, each induced subgraph G_i is connected and satisfies

$$\operatorname{diam}(G_i) \leq 2K.$$

The first result follows from maximality: if a vertex were more than K hops from every centroid, it could itself be added to the K-MIS. For the second result, consider a shortest path from v to its assigned centroid s_i. If an intermediate vertex on this path were assigned to another centroid, that centroid would also be preferred by v, contradicting the nearest-centroid assignment. Therefore, the path remains within V_i. This property is particularly useful for our reconstruction framework. It guarantees that each supernode represents a connected local region centred around s_i, with radius at most K. It also motivates the use of centroid-relative structural information introduced in Section IV-C. K-MIS additionally provides a reduction in the number of graph units processed at the coarse level. Let

$$q = \left\lfloor \frac{K}{2} \right\rfloor.$$

Since two K-MIS centroids have shortest-path distance greater than K, their radius-q neighbourhoods cannot overlap.

Proposition 2 (Bound on the number of supernodes)**.** *Let $M = |S|$ denote the number of K-MIS centroids. The radius-q neighbourhoods centred at S are pairwise disjoint, and therefore*

$$\sum_{i=1}^{M} |B_G(s_i, q)| \leq n.$$

For a connected graph with a non-trivial partition $M > 1$, each such neighbourhood contains at least $q + 1$ vertices, giving

$$M \leq \frac{n}{q+1} = \frac{n}{\lfloor K/2 \rfloor + 1}.$$

To see why the neighbourhoods are disjoint, suppose that a vertex belonged to both $B_G(s_i, q)$ and $B_G(s_j, q)$. By the triangle inequality,

$$d_G(s_i, s_j) \leq 2q \leq K,$$

which contradicts the K-MIS separation condition.

The bound is a worst-case guarantee rather than an estimate of the actual number of supernodes. In practice, the local neighbourhoods around selected centroids may contain

substantially more than $q+1$ vertices, leading to a stronger reduction. We therefore also report the empirical ratio M/n and the resulting supernode-size distribution in Section VI. These properties provide the structural basis for the computational analysis in Section V: instead of processing the original graph as a single n-node object, the diffusion models operate on a collection of smaller K-local regions and their connected pairs.

C. Structure-Aware Graph Synopsis

Although K-MIS determines the partition and coarse connectivity, the coarse edge counts alone do not specify the fine-grained arrangement of edges. For example, two supernodes can have the same size and internal edge count while exhibiting very different local topologies. Similarly, a pair of supernodes can contain the same number of cross edges while distributing those edges very differently among their vertices. We therefore augment the basic coarse representation with structural summaries for both intra- and inter-supernode reconstruction. Intra-supernode structure For a supernode V_i, we retain its size n_i and internal density

$$\rho_i = \frac{2m_i}{n_i(n_i-1)},$$

for $n_i > 1$, with $\rho_i = 0$ for a singleton supernode. To capture information that is not represented by size and density alone, we additionally define a centroid-relative hop distribution. For $\ell = 1, \dots, K$,

$$h_i(\ell) = \frac{|\{v \in V_i \setminus \{s_i\} : d_{G_i}(s_i, v) = \ell\}|}{n_i - 1}.$$

The vector

$$h_i = [h_i(1), \dots, h_i(K)]$$

describes how the vertices of V_i are distributed around its centroid. Proposition 1 ensures that all vertices lie within K hops of s_i and that their shortest paths to the centroid remain inside G_i. Therefore, h_i provides a compact description of the radial organisation of the supernode. The intra-supernode structural condition is consequently represented by

$$c_i = [n_i, \rho_i, K, h_i].$$

Inter-supernode structure For two connected supernodes V_i and V_j, the coarse weight w_{ij} records the total number of cross edges. However, w_{ij} alone does not indicate whether these edges are concentrated on a small number of vertices or distributed broadly across the two supernodes. For each vertex $u \in V_i$, we therefore define its cross-degree towards V_j as

$$d_{i\to j}(u) = |\{v \in V_j : (u, v) \in E\}|.$$

We summarise these values using a normalised cross-degree histogram $r_{i\to j}$. In our implementation, cross-degrees are divided into five bins,

$$0,\ 1,\ 2,\ 3,\ \geq 4.$$

For a bin b, $r_{i\to j}(b)$ denotes the proportion of vertices in V_i whose cross-degree towards V_j belongs to that bin. We similarly define $r_{j\to i}$ from the opposite side. The two directions are retained separately because they characterise different endpoint distributions. For example, the same w_{ij} may correspond to a structure where many vertices in V_i connect to a small number of vertices in V_j, or to a structure in which the edges are more evenly distributed across both supernodes. The pair-level structural information is therefore represented by

$$c_{ij} = [c_i, c_j, w_{ij}, r_{i\to j}, r_{j\to i}].$$

Collectively, we denote the resulting structure-aware graph synopsis by

$$\mathcal{S}_K(G) = \left(G_c,\ \{c_i\}_{i=1}^{M},\ \{c_{ij}\}_{(i,j)\in E_c}\right).$$

This synopsis serves as the conditioning information for the two diffusion models introduced in Section V.

Both summaries are invariant to vertex relabelling: h_i depends only on the centroid-relative distance distribution, while $r_{i\to j}$ depends only on the distribution of cross-degrees. They therefore capture structural information without relying on arbitrary vertex ordering.

D. Reconstruction Ambiguity

The motivation for retaining h_i and $r_{i\to j}$ can also be understood from the number of possible fine-grained graphs that are compatible with a coarse representation. Consider first an intra-supernode graph containing n_i labelled vertices and m_i edges. If only n_i and m_i are known, the number of possible simple graphs is

$$|\Omega_i^{\text{base}}| = \binom{\binom{n_i}{2}}{m_i}.$$

Thus, even when size and density are known exactly, a potentially large number of different internal topologies remain feasible. The hop distribution h_i introduces additional constraints. Let

$$a_\ell = (n_i - 1)h_i(\ell)$$

denote the number of non-centroid vertices at distance ℓ from s_i, with $a_0 = 1$. For a fixed assignment of labelled vertices to these hop layers, an edge can occur only within the same layer or between two consecutive layers; moreover, every vertex in layer $\ell > 0$ must have at least one neighbour in layer $\ell - 1$. These conditions ensure that its shortest-path distance from the centroid is exactly ℓ. For a fixed layer assignment, the generating function for all graphs consistent with these hop distances is

$$F_i(x) = (1+x)^{\sum_{\ell=1}^{K}\binom{a_\ell}{2}} \prod_{\ell=1}^{K} \left((1+x)^{a_{\ell-1}} - 1\right)^{a_\ell}.$$

The coefficient of x^{m_i} gives the number of graphs with exactly m_i edges that are consistent with the specified hop-layer structure. If only the layer sizes are retained rather than the assignment of individual vertices to layers, this quantity is additionally multiplied by

$$\frac{(n_i-1)!}{\prod_{\ell=1}^{K} a_\ell!}.$$

Algorithm 1 Capacity-Aware Balanced K-MIS (CB-KMIS)

Require: Graph $G = (V, E)$, hop bound k, capacity b
Ensure: Centroids S and partition $\mathcal{C}$

for all $v \in V$ **in parallel do**
  $q(v) \leftarrow |B_k(v)|$
  $\phi(v) \leftarrow \min\left\{\frac{q(v)}{b}, \frac{b}{q(v)}\right\}$
end for
$S_0 \leftarrow \mathrm{KMIS}(G, k, \phi)$
$\mathcal{R} \leftarrow \textsc{NearestAssign}(G, S_0, k)$
$S \leftarrow \emptyset, \quad \mathcal{C} \leftarrow \emptyset$
for all $(c_i, R_i) \in (S_0, \mathcal{R})$ **in parallel do**
  if $|R_i| \le b$ **then**
    $S \leftarrow S \cup \{c_i\}$, $\mathcal{C} \leftarrow \mathcal{C} \cup \{R_i\}$
  else
    $S_i \leftarrow \mathrm{LBSPR}(G[R_i], c_i, k, b)$
    repeat
      $(\mathcal{C}_i, U_i) \leftarrow \textsc{CapBFS}(G[R_i], S_i, k, b)$
      for all connected components H of $G[U_i]$ **do**
        $S_i \leftarrow S_i \cup \{\textsc{ChooseCentroid}(H)\}$
      end for
    until $U_i = \emptyset$
    $S \leftarrow S \cup S_i$, $\mathcal{C} \leftarrow \mathcal{C} \cup \mathcal{C}_i$
  end if
end for
return $(S, \mathcal{C})$ =0

Algorithm 2 Load-Balanced Shortest-Path Refinement (LB-SPR)

Require: Oversized region $G_i = G[R_i]$, centroid c_i, hop bound k, capacity b
Ensure: Refined centroid set S_i

Compute BFS distance $d(v) = d_{G_i}(c_i, v)$
$P(v) \leftarrow \{u \in N(v) : d(u) = d(v) - 1\}$
// Build load-balanced shortest-path tree
for all $v \in R_i$ **do**
  $w(v) \leftarrow 1$
end for
for $r = k$ **downto** 1 **do**
  for all $v : d(v) = r$ in decreasing $w(v)$ **do**
    $p^* \leftarrow \arg\min_{p \in P(v)} (\max\{0, w(p) + w(v) - b\}, w(p) + w(v))$
    $\mathrm{parent}(v) \leftarrow p^*$
    $w(p^*) \leftarrow w(p^*) + w(v)$
  end for
end for
Compute subtree sizes $s_T(v)$
$\tilde{s}_T(v) \leftarrow s_T(v)$
$q \leftarrow \lceil |R_i|/b \rceil$
$S_i \leftarrow \{c_i\}$
// Select additional centroids
while $|S_i| < q$ **do**
  for all unselected v **do**
    $\psi(v) \leftarrow \min\left\{\frac{\tilde{s}_T(v)}{b}, \frac{b}{\tilde{s}_T(v)}\right\}$
  end for
  $v^* \leftarrow \arg\max_v \psi(v)$
  $m \leftarrow \tilde{s}_T(v^*)$
  $S_i \leftarrow S_i \cup \{v^*\}$
  for all $u \in \mathrm{Ancestors}(v^*)$ **do**
    $\tilde{s}_T(u) \leftarrow \max\{0, \tilde{s}_T(u) - m\}$
  end for
  Mark v^* as selected
end while
return S_i =0

This characterisation shows that h_i restricts the candidate reconstruction space beyond that defined by n_i and m_i alone. A similar ambiguity arises between supernodes. Given only the sizes n_i and n_j and the cross-edge count w_{ij}, there are

$$|\Omega_{ij}^{\mathrm{base}}| = \binom{n_i n_j}{w_{ij}}$$

possible bipartite adjacency blocks. The cross-degree summaries $r_{i\to j}$ and $r_{j\to i}$ restrict this space by constraining how the w_{ij} edges can be distributed among vertices on both sides. Because our summaries use a binned ≥ 4 category rather than an exact degree sequence, they do not uniquely determine the cross-edge structure. Instead, they retain a compact intermediate representation between a single edge count and the full bipartite adjacency matrix. Therefore,

$$\Omega_{ij}^{r} \subseteq \Omega_{ij}^{\mathrm{base}},$$

where Ω_{ij}^{r} denotes the set of bipartite graphs consistent with the retained cross-degree summaries. The purpose of h_i and $r_{i\to j}$ is therefore not to uniquely determine the original graph, but to reduce the set of plausible fine-grained reconstructions while maintaining a compact coarse representation. Whether this reduction in structural ambiguity leads to improved generation fidelity is evaluated empirically through the conditioning ablations in Section VI.

V. Coarse-to-Fine Graph Diffusion

- Prove permutation invariance of h and r

Given the structure-aware coarse representation introduced in Section IV, we reconstruct the fine-grained graph through two conditional discrete diffusion models. The first model, $\mathcal{D}_{\mathrm{intra}}$, reconstructs the internal topology of each supernode, while the second model, $\mathcal{D}_{\mathrm{inter}}$, reconstructs the edges between connected supernodes. The generated intra- and inter-supernode structures are then assembled to form the final graph. This decomposition avoids applying diffusion directly to the full graph and instead performs generation over a collection of smaller local blocks defined by the K-MIS partition.

A. Intra-Supernode diffusion

For each supernode V_i, let $G_i = G[V_i]$ denote its induced subgraph and let $L_i \in \{0, 1\}^{n_i \times n_i}$ denote its adjacency matrix. The objective of D_{intra} is to generate an internal graph $\widehat{G}_i$ whose structure is consistent with the information retained for supernode i. We build D_{intra} upon the discrete graph diffusion framework of DiGress. During training, the clean

Algorithm 3 Capacity-Bounded Multi-Source BFS (CapBFS)

Require: Region $G_i = (V_i, E_i)$, centroids S_i, hop bound k, capacity b
Ensure: Connected blocks $\mathcal{C}_i$ and residual nodes U_i

```
for all c_j ∈ S_i do
    C_j ← {c_j}
end for
U_i ← V_i \ S_i
for r = 1 to k do
    for all v ∈ U_i reached at level r do
        A(v) ← {c_j : ∃u ∈ C_j, (u, v) ∈ E_i, |C_j| < b}
        if A(v) ≠ ∅ then
            c* ← arg min_{c_j ∈ A(v)} |C_j|
            C_{c*} ← C_{c*} ∪ {v}
            U_i ← U_i \ {v}
        end if
    end for
end for
return ({C_j}_{c_j ∈ S_i}, U_i) =0
```

adjacency matrix L_i is progressively corrupted through a categorical forward diffusion process. Let L_i^t denote the noisy adjacency matrix at diffusion step t. The reverse model learns to recover the clean edge distribution from L_i^t, conditioned on the structural information associated with the supernode.

The conditioning vector is defined as

$$c_i = [n_i, \rho_i, K, h_i],$$

where n_i denotes the number of vertices, ρ_i denotes the internal edge density, K is the coarsening parameter, and h_i describes the hop-distance distribution around the centroid. The graph size n_i determines the number of vertices generated for the supernode, while ρ_i and h_i provide additional structural information to guide edge generation. Because h_i is defined relative to the centroid, the centroid must remain identifiable during generation. We therefore introduce a fixed centroid indicator as a node feature: the centroid node is assigned value one and all remaining nodes are assigned value zero. These node features remain unchanged throughout the diffusion process, and diffusion is applied only to the edge states. The condition c_i is supplied through the graph-level conditioning channel and remains unchanged during denoising. The reverse process can therefore be written as

$$p_\theta(L_i^{t-1} \mid L_i^t, c_i),$$

where θ denotes the parameters of D_{intra}. Starting from a noisy edge configuration, the reverse diffusion process generates an adjacency matrix $\widehat{L}_i$, from which the reconstructed supernode $\widehat{G}_i$ is obtained. The structural statistics in c_i are used as conditioning information rather than hard constraints. Consequently, the model is encouraged to generate graphs with structures compatible with the supplied conditions, but exact reproduction of ρ_i or h_i is not explicitly enforced.

B. Inter-Supernode diffusion

After reconstructing the internal topology of the supernodes, we generate the edges between connected supernode pairs. For a pair $(i, j) \in E_c$, let

$$B_{ij} \in \{0, 1\}^{n_i \times n_j}$$

denote the bipartite adjacency block containing the edges between V_i and V_j. Unlike D_{intra}, which generates the entire internal adjacency matrix, D_{inter} diffuses and predicts only the cross-supernode edges.

Instead of using the complete internal adjacency matrices L_i and L_j, we represent each supernode using the internal degree of its nodes. For a node $v \in V_i$, its normalised internal degree is

$$x_v = \frac{d_i^{\text{in}}(v)}{n_i - 1},$$

where $d_i^{\text{in}}(v)$ is the number of neighbours of v within the same supernode. We collect these node features as

$$X_i = [x_v]_{v \in V_i}, \qquad X_j = [x_u]_{u \in V_j}.$$

Thus, X_i and X_j provide simple node-level information about the internal connectivity of the two supernodes without requiring their full adjacency matrices.

At diffusion step t, D_{inter} takes the node features X_i and X_j, together with the corrupted cross-edge block B_{ij}^t, as input. Only the cross-edge block is diffused, while X_i and X_j remain fixed throughout the diffusion process.

The pair-level condition is defined as

$$c_{ij} = [n_i, n_j, w_{ij}, r_{i \to j}, r_{j \to i}],$$

where w_{ij} denotes the total number of edges between the two supernodes, and $r_{i \to j}$ and $r_{j \to i}$ describe the corresponding cross-degree distributions. The cross-edge count captures the overall interaction strength between the two supernodes, while the cross-degree distributions provide additional information about how these edges are distributed among their vertices. The reverse process is modelled as

$$p_\phi \left(B_{ij}^{t-1} \mid B_{ij}^t, X_i, X_j, c_{ij} \right),$$

where ϕ denotes the parameters of D_{inter}. Since the internal structures serve only as context, the training loss is evaluated exclusively on the cross-edge entries. During training, the node-level structural features X_i and X_j are computed from the ground-truth internal structures L_i and L_j. At inference time, the ground-truth internal structures are unavailable. We therefore compute the corresponding features from the generated intra-supernode structures $\hat{L}_i$ and $\hat{L}_j$, resulting in $\hat{X}_i$ and $\hat{X}j$. These generated node-level features are then used as the structural context for $\mathcal{D}_{\text{inter}}$. Hence, D_{inter} is trained using ground-truth internal structures but operates on generated internal structures during inference. This introduces a train–inference distribution shift, whose effect is evaluated experimentally in Section VI.

C. Whole-graph synthesis

Given the coarse representation of a target graph, reconstruction proceeds in two stages. First, D_{intra} independently generates an internal adjacency matrix $\widehat{L}_i$ for each of the M target supernodes. This produces a collection of generated local graphs

$$\{\widehat{G}_1, \ldots, \widehat{G}_M\}.$$

Second, for every coarse edge $(i, j) \in E_c$, D_{inter} generates the corresponding cross-edge block $\widehat{B}_{ij}$ using the generated internal structures and the pair-level structural condition c_{ij}. The complete adjacency matrix $\widehat{A}$ is obtained by placing each $\widehat{L}_i$ on the corresponding diagonal block and each $\widehat{B}_{ij}$ on its corresponding off-diagonal block:

$$\widehat{A}_{ij} = \begin{cases} \widehat{L}_i, & i = j, \\ \widehat{B}_{ij}, & (i, j) \in E_c, \\ 0, & \text{otherwise.} \end{cases}$$

Cross edges are therefore generated only between supernode pairs connected in the coarse graph. The resulting graph is denoted by $\widehat{G}$. Because multiple fine-grained graphs can share the same coarse representation, the objective is not exact recovery of the original edge set. Instead, the framework generates a plausible fine-grained realisation that is consistent with the retained coarse information and preserves important structural characteristics of the original graph.

D. Complexity and implementation

The coarse-to-fine decomposition also reduces the size of the graph states processed by the diffusion models. A direct dense graph diffusion model represents up to $O(n^2)$ possible edge states for a graph with n vertices. In contrast, our framework performs diffusion over smaller intra- and inter-supernode blocks. For D_{intra}, the total number of adjacency entries processed in one diffusion step is proportional to

$$\sum_{i=1}^{M} n_i^2,$$

where $\sum_i n_i = n$. Let

$$b = \max_i n_i$$

denote the maximum supernode size. Then

$$\sum_{i=1}^{M} n_i^2 \le b \sum_{i=1}^{M} n_i = bn.$$

Therefore, the edge-state space considered by D_{intra} is bounded by $O(nb)$, which is substantially smaller than $O(n^2)$ when the supernodes remain small relative to the original graph. For D_{inter}, cross-edge prediction is performed only for supernode pairs connected in the coarse graph. The total number of predicted cross-edge states is

$$\sum_{(i,j)\in E_c} n_i n_j.$$

Let $m_c = |E_c|$ denote the number of coarse edges. Since $n_i, n_j \le b$,

$$\sum_{(i,j)\in E_c} n_i n_j \le m_c b^2.$$

Thus, the number of cross-edge states considered by D_{inter} is bounded by $O(m_c b^2)$. In our implementation, D_{inter} processes the complete pair-level graph containing both internal and cross-supernode blocks. Hence, the dense state size processed for one connected pair is proportional to $(n_i + n_j)^2$. Across all coarse edges,

$$\sum_{(i,j)\in E_c} (n_i + n_j)^2 \le 4 m_c b^2.$$

For T reverse diffusion steps, these state-size costs scale linearly with T. The resulting computational benefit therefore depends on both the maximum supernode size b and the sparsity of the coarse graph G_c. Together with the bound on the number of K-MIS supernodes established in Section IV, this analysis shows that K-MIS decomposes full-graph generation into a collection of smaller local generation problems. In Section VI, we report the observed supernode sizes, coarse-node and coarse-edge reduction ratios, and runtime to quantify this reduction empirically.

VI. Experiments

We evaluate the proposed framework through five studies: CB-KMIS coarsening, intra/inter-structure reconstruction, synopsis effect, cross-graph transfer, and full-graph quality, and computational scalability. The first study examines whether CB-KMIS produces suitable reconstruction units. The remaining studies assess local and global structural fidelity, the effects of structural conditioning and target adaptation, and computational cost.

A. Experimental Setup

1) Datasets: We consider road, collaboration, and citation networks to examine reconstruction across graphs with different sizes and connectivity patterns. Road networks provide sparse graphs with low vertex degrees, while collaboration and citation networks extend the evaluation to more heterogeneous connectivity patterns. Table II summarises the dataset characteristics.

The source-to-target settings are LUXEMBOURG→OLDENBURG, DBLP→OGBL-COLLAB, and CITATION-NETWORK-V19→ OBGN-ARXIV. These settings examine transfer within each domain while allowing the source and target graphs to differ in size and structure. Our framework operates on undirected, unweighted, simple graphs.

2) Baselines: We distinguish coarsening baselines, which evaluate the partition, from reconstruction and generation baselines, which evaluate the resulting graph structure.

TABLE II
CHARACTERISTICS OF THE GRAPH DATASETS. A DASH INDICATES AN UNREPORTED STATISTIC.

Dataset	$\|V\|$	$\|E\|$	Density	Min Deg.	Max Deg.	Mean Deg.	Median Deg.	Type
LUXEMBOURG	114,599	119,666	1.82e-05	1	6	2.09	2	Road Network
OLDENBURG [3]	6,105	7,029	3.77e-04	1	5	2.30	2	Road Network
VERMONT	97,975	107,558	–	1	–	–	–	Road Network
DBLP [15]	317,080	1,049,866	2.09e-05	1	343	6.62	4	Collaboration
OGBL-COLLAB	235,868	1,285,465	–	1	382	8.2	4	Collaboration
CITATION-NETWORK-V19	7,289,177	85,014,559	–	–	–	–	–	Citation
OBGN-ARXIV [10]	169,343	1,166,243	–	–	–	–	–	Citation

a) Graph coarsening.: We compare CB-KMIS with Random-Centroid, random edge contraction (REC), Variation Neighbourhoods, Leiden, and original K-MIS. Random-Centroid samples M vertices uniformly as centroids and assigns each remaining vertex to its nearest centroid by shortest-path distance, with ties resolved by a fixed ordering. The partition is converted to a weighted coarse graph in the same way as K-MIS. REC provides an edge-contraction baseline, Variation Neighbourhoods represents spectral coarsening, and Leiden provides a community-based partition. Original K-MIS is the main reference for evaluating the capacity-aware selection and refinement introduced by CB-KMIS.

We compare methods at matched or closely matched reduction ratios and report the achieved ratio for each method. Method-specific parameters are selected using `[matching procedure and tolerance]`. We also report maximum supernode size and coarse-edge count, since equal reduction ratios do not imply equal reconstruction workloads.

b) Non-learning reconstruction.: We include two random reconstruction baselines. The first, denoted ER, uses Erdős–Rényi $G(n, m)$ sampling [7]. For each supernode, it samples exactly m_i internal edges uniformly from the $\binom{n_i}{2}$ possible pairs. For each connected supernode pair (i, j), it samples exactly w_{ij} cross edges uniformly from the $n_i n_j$ possible bipartite pairs. The blocks are assembled according to the coarse graph. ER preserves n_i, m_i, and w_{ij} but does not use the radial or cross-degree summaries.

The second baseline, Synopsis-Constrained Random Graph (SC-RG), is inspired by degree-preserving configuration models [8], [21]. For intra-supernode reconstruction, $\mathbf{h}_i$ defines centroid-based hop layers, and edges are randomly placed subject to the distance constraints and prescribed edge count. For inter-supernode reconstruction, the directional summaries $\mathbf{r}_{i\to j}$ and $\mathbf{r}_{j\to i}$ are converted into degree constraints, and a random bipartite graph is constructed while preserving w_{ij}. This baseline examines whether learned reconstruction improves on random edge placement under the retained structural information.

c) Learned graph generation.: We include DiGress [27] as a discrete diffusion baseline for local generation. For full-graph comparisons, we consider iterative local expansion (ILE) [2], which generates graphs through local expansion and refinement; HiGen [14], which models hierarchical generation over communities; and HiGGs [6], which provides a hierarchical graph-generation framework.

We distinguish methods that receive the target synopsis from methods that receive only the target graph size or their native inputs. ER and SC-RG provide synopsis-conditioned references. Comparisons with other generators are interpreted according to their stated information access, since target summaries provide structural information beyond graph size.

3) Evaluation Metrics: We evaluate partition quality, local and global structural fidelity, and computational cost. Smaller discrepancy values indicate closer agreement with the reference graph.

a) Coarsening quality.: Let n be the number of vertices in the original graph, n_i and m_i the numbers of vertices and internal edges in supernode i, and M the number of supernodes. Let E_c contain unique unordered pairs of distinct connected supernodes, so $|E_c|$ is the number of superedges. The reduction ratio is

$$r = 1 - \frac{M}{n}. \tag{1}$$

We report the mean, median, maximum, and 95th percentile of supernode sizes. The 95th percentile, $\mathrm{P95}(n_i)$, characterises the upper range of typical sizes, while the maximum identifies extreme reconstruction units.

Following the convention used in the coarsening table, we report the reciprocal coefficient of variation:

$$\mathrm{CV}(n_i) = \frac{\sigma_n}{\bar{n}}, \qquad \bar{n} = \frac{n}{M}, \tag{2}$$

where σ_n is the population standard deviation of supernode sizes, and smaller values indicate more balanced partitions.

Average radius (Avg. Rad.) measures locality. For a connected supernode $G_i = G[V_i]$, its radius is

$$\mathrm{rad}(G_i) = \min_{v\in V_i} \max_{u\in V_i} d_{G_i}(v, u), \tag{3}$$

where distances are measured within the induced subgraph. Smaller average radii indicate more compact local regions.

We quantify adjacency workloads using

$$C_{\mathrm{intra}} = \sum_{i=1}^{M} n_i^2, \qquad C_{\mathrm{inter}} = \sum_{(i,j)\in E_c} n_i n_j. \tag{4}$$

These measure the intra-supernode adjacency space and candidate cross-edge space, respectively, counting each coarse pair

once. Their normalised values, C_{intra}/n^2 and C_{inter}/n^2, support comparisons across graph sizes. Since the inter-supernode implementation processes complete pair-level graphs, its workload can also be characterised by

$$C_{\text{pair}} = \sum_{(i,j)\in E_c} (n_i + n_j)^2. \tag{5}$$

This quantity accounts for repeated processing of supernode context across coarse edges. Workload measures are interpreted alongside measured runtime and memory.

Restricted spectral approximation error provides a secondary measure of structural preservation. Let L be the original Laplacian and $\widetilde{L}$ the coarse Laplacian lifted to the original vertex space using the evaluation operators. For each selected non-trivial eigenvector u_ℓ with $u_\ell^\top L u_\ell > 0$, we compute

$$\epsilon_\ell = \frac{\left| u_\ell^\top \widetilde{L} u_\ell - u_\ell^\top L u_\ell \right|}{u_\ell^\top L u_\ell}. \tag{6}$$

Smaller values indicate better preservation of Laplacian energy along the evaluated directions. Our main coarsening evaluation concerns suitability for generative reconstruction.

b) Intra-supernode reconstruction.: Degree W1 is the 1-Wasserstein distance between the empirical degree distributions pooled across generated and reference test supernodes, yielding one score per configuration.

Shortest-path W1 measures the discrepancy in internal distance structure. For each supernode, we compute shortest-path distances between distinct vertex pairs within its largest connected component. We then calculate the 1-Wasserstein distance between the pooled distance distributions of generated and reference test supernodes. Vertices outside the largest connected component do not contribute to this metric.

Radial fidelity is measured by

$$E_h = \frac{1}{N} \sum_{i=1}^{N} D_{\text{JS}} \left(\mathbf{h}_i, \widehat{\mathbf{h}}_i \right), \tag{7}$$

where N is the number of evaluated supernodes, $\mathbf{h}_i$ and $\widehat{\mathbf{h}}_i$ are the reference and generated centroid-relative hop distributions, respectively, and D_{JS} denotes Jensen–Shannon divergence.

For spectral discrepancy, we compute the median 1-Wasserstein distance between each generated supernode's Laplacian spectrum and those of all reference test supernodes. The reported score is the median of these values across generated supernodes. This metric measures resemblance to the reference collection rather than reconstruction of the specific conditioned supernode. Lower values indicate better agreement for all metrics.

c) Inter-supernode reconstruction.: We evaluate cross-degree distributions using 1-Wasserstein distance and agreement with the directional summaries using $E_{r_{i\leftrightarrow j}}$. Cross-edge count consistency is assessed separately. Metrics preserved exactly by construction are interpreted as constraint satisfaction rather than independent evidence of fidelity.

d) Full-graph quality.: We evaluate assembled graphs using degree-distribution W1, relative errors in average shortest-path length (APL) and global clustering coefficient (GCC), and clustering, orbit, and spectral discrepancies.

Clustering discrepancy compares the distributions of local clustering coefficients, measuring how closely the model preserves triangle formation around individual vertices. For a vertex v with degree d_v and t_v incident triangles, its local clustering coefficient is

$$c_v = \begin{cases} \dfrac{2t_v}{d_v(d_v-1)}, & d_v \geq 2, \\ 0, & d_v < 2. \end{cases} \tag{8}$$

Unlike GCC, which summarises triangle closure across the whole graph, this distribution captures variation in local clustering among vertices.

Orbit discrepancy compares graphlet-orbit statistics, which count how often vertices occupy distinct structural roles in small induced subgraphs. It assesses local connection patterns beyond vertex degrees and triangle formation.

Local measurements use held-out reconstruction instances, whereas full-graph measurements use the complete generated graph. We report these separately because local fidelity does not necessarily imply accurate global structure. Lower discrepancies indicate better agreement.

e) Computational efficiency.: We report coarsening, synopsis extraction, training, and generation times, together with peak CPU and GPU memory where applicable. Generation time refers to one complete graph, with the included stages stated explicitly. We distinguish successful runs, memory failures, timeouts, and untested settings.

4) Implementation Details:

a) Model configuration.: The intra-supernode model builds on discrete graph diffusion [27] and uses $\mathbf{c}_i = [n_i, \rho_i, K, \mathbf{h}_i]$. A fixed node indicator identifies the centroid, and diffusion is applied to edge states. The inter-supernode model diffuses only the cross-edge block, conditioned on $\mathbf{c}_{ij} = [n_i, n_j, w_{ij}, \mathbf{r}_{i\to j}, \mathbf{r}_{j\to i}]$. Its node-level context consists of normalised internal degrees computed from real supernodes during training and generated supernodes during inference.

The CB-KMIS hop parameter and capacity are $K =$ `[value]` and $b =$ `[value]`. The diffusion models use `[architecture, depth, hidden dimensions, attention heads, and diffusion schedule]`. Training uses `[optimiser]`, learning rate `[value]`, batch size `[value]`, and `[number]` epochs. Checkpoints are selected using `[validation criterion]`.

b) Training and transfer protocol.: Let G_s and G_t denote the source and target graphs. Let L_i be the real internal adjacency matrix of supernode i, and $\widehat{L}_i$ its generated counterpart. The target supernodes at hop parameter K form a set $\mathcal{M}_K$, partitioned into a fine-tuning set $\mathcal{A}$ and a held-out set $\mathcal{H}$:

$$\mathcal{A} \cup \mathcal{H} = \mathcal{M}_K, \qquad \mathcal{A} \cap \mathcal{H} = \varnothing. \tag{9}$$

The current target split assigns 80% of supernodes to fine-tuning and 20% to held-out evaluation. We distinguish three configurations:

- **Configuration A: source evaluation.** Models are trained on G_s and evaluated on held-out source instances to assess reconstruction within the training domain.
- **Configuration B: source-only transfer (ZS).** Models are trained only on G_s. No target fine-grained examples are used for model optimisation. At inference, the full target synopsis provides the conditions for generating all target blocks. Local evaluation uses $\mathcal{H}$.
- **Configuration C: target adaptation (FT).** The same source-trained models are fine-tuned on $\mathcal{A}$ and evaluated on the same $\mathcal{H}$ used in Configuration B. Full-graph generation uses the target synopsis as in Configuration B. Target-only models use the same target training subset without source pre-training.

For full-graph synthesis, all intra-supernode and cross-edge blocks are generated before assembly. Training blocks are not copied into the generated graph. Target synopsis access is reported separately from access to target training examples.

c) Ablations and reproducibility.: The radial and cross-degree ablations compare models with and without the corresponding summaries, using matching splits and training budgets while holding other settings fixed. Experiments use `[CPU model and core count]`, `[system RAM]`, and `[GPU model and memory]`. The software environment comprises `[operating system, Python, PyTorch, CUDA, and relevant package versions]`. Results are reported over `[number]` independent training runs and `[number]` generated samples per setting. Uncertainty denotes `[standard deviation or confidence interval]`.

B. Evaluation of CB-KMIS Coarsening

a) Objective and design.: We examine whether CB-KMIS produces connected, size-controlled supernodes while maintaining structural fidelity and manageable generation workloads. Table III compares the methods at comparable reduction ratios. We assess balance and maximum size together with coarse-edge count, since smaller supernodes may increase the number of cross-supernode reconstruction tasks.

C. Intra-Supernode Reconstruction

a) Objective and design.: We assess whether D_{intra} recovers local topology beyond the supplied size and density, and whether radial conditioning improves reconstruction. Table IV compares local fidelity, while Table **??**(a) isolates the effect of $\mathbf{h}_i$.

TABLE IV
INTRA-SUPERNODE RECONSTRUCTION: (A) COMPARISON WITH BASELINES AND (B) EFFECT OF RADIAL CONDITIONING. LOWER IS BETTER.

Target	Method	Degree W1 ↓	Shortest-path W1 ↓	E_h ↓
(a) Baseline comparison				
OLDENBURG	Ours	0.04 ± 0.012	–	0.286 ± 0.022
	ER	0.46 ± 0.017	–	0.581 ± 0.032
	SC-RG	0.20 ± 0.009	–	0
	DiGress	$\mathbf{0.03 \pm 0.014}$	–	–
OGBL-COLLAB	Ours	–	–	–
	ER	–	–	–
	SC-RG	–	–	–
	DiGress	–	–	–
OBGN-ARXIV	Ours	–	–	–
	ER	–	–	–
	SC-RG	–	–	–
	DiGress	–	–	–
(b) Radial-conditioning ablation				
OLDENBURG	w/o $\mathbf{h}_i$	0.0517 ± 0.0098	–	0.362 ± 0.0104
	$+\mathbf{h}_i$	0.0434 ± 0.0115	–	0.286 ± 0.0218
OGBL-COLLAB	w/o $\mathbf{h}_i$	–	–	–
	$+\mathbf{h}_i$	–	–	–
OBGN-ARXIV	w/o $\mathbf{h}_i$	–	–	–
	$+\mathbf{h}_i$	–	–	–

On OLDENBURG, the proposed model has lower mean degree discrepancy than ER and SC-RG, while DiGress has the lowest reported mean. SC-RG matches the radial summary exactly by construction but has a larger degree discrepancy than the learned models. This shows why summary agreement should be considered alongside structural properties that are not enforced directly.

b) Effect of radial conditioning.: Adding $\mathbf{h}_i$ lowers both the mean radial discrepancy and degree discrepancy in Table **??**(a). These results support radial conditioning for the evaluated Oldenburg setting; they do not establish improvement across all domains or structural metrics.

D. Inter-Supernode Reconstruction

a) Objective and design.: We assess whether D_{inter} recovers the distribution of cross edges among vertices and whether directional cross-degree conditioning improves this distribution. Table V compares the reconstruction models; Table **??**(b) isolates the additional summaries.

TABLE III
COARSENING COMPARISON AT COMPARABLE REDUCTION RATIOS. ARROWS INDICATE THE PREFERRED DIRECTION. CV FOLLOWS EQUATION 2.

Method	Type	r	$\lvert E_c \rvert$	CV(n_i)↓	P95(n_i)	Max n_i	Median n_i	Mean n_i	Avg. Rad.↓	C_{intra} ↓	C_{inter} ↓	ϵ ↓
Random-Centroid	Random	–	–	–	–	–	–	–	–	–	–	–
REC	Edge Contraction	–	–	–	–	–	–	–	–	–	–	–
Variation N.	Spectral	–	–	–	–	–	–	–	–	–	–	–
Leiden	Community	–	–	–	–	–	–	–	–	–	–	–
K-MIS	Locality	–	–	–	–	–	–	–	–	–	–	–
CB-KMIS	Capacity-Aware	–	–	–	–	–	–	–	–	–	–	–

TABLE V
INTER-SUPERNODE RECONSTRUCTION: (A) COMPARISON WITH BASELINES AND (B) EFFECT OF DIRECTIONAL CROSS-DEGREE CONDITIONING. LOWER IS BETTER.

Target	Method	Cross-degree W1 ↓	$E_{r_{i\leftrightarrow j}}$ ↓
(a) Baseline comparison			
OLDENBURG	Ours	0.0008 ± 0.0003	0.017 ± 0.0045
	ER	0.0020 ± 0.0004	0.012 ± 0.0009
	SC-RG	0.0011 ± 0.0005	0
OGBL-COLLAB	Ours	–	–
	ER	–	–
	SC-RG	–	–
OBGN-ARXIV	Ours	–	–
	ER	–	–
	SC-RG	–	–
(b) Cross-degree-conditioning ablation			
OLDENBURG	w_{ij} only	0.00227 ± 0.000337	0.0163 ± 0.0028
	$+\mathbf{r}_{i\leftrightarrow j}$	0.00083 ± 0.00028	0.0171 ± 0.0045
OGBL-COLLAB	w_{ij} only	–	–
	$+\mathbf{r}_{i\leftrightarrow j}$	–	–
OBGN-ARXIV	w_{ij} only	–	–
	$+\mathbf{r}_{i\leftrightarrow j}$	–	–

On OLDENBURG, the proposed model has the lowest reported mean cross-degree W1 among the three reconstruction methods. However, ER has a lower mean summary discrepancy, while SC-RG matches that summary exactly. The metrics therefore give different rankings, which must be interpreted according to their binning and aggregation rules.

b) Effect of cross-degree conditioning.: The additional summaries reduce mean cross-degree W1 in Table **??**(b), but do not reduce the reported summary discrepancy. The evidence supports improved cross-degree fidelity in this setting, rather than improvement on every measure.

E. Full-Graph Quality and Cross-Graph Transfer

a) Objective and design.: We examine whether local reconstruction yields faithful full-graph structure and whether target adaptation improves source-to-target transfer. Table VI compares training configurations, and Table VII compares full-graph synthesis methods.

b) Source-only transfer.: For Configuration B, competing generators are trained on G_s and asked to generate a graph with the target size. For example, models trained on LUXEMBOURG generate a graph with $6{,}105$ vertices for comparison with OLDENBURG. This adapts the competing methods to the transfer setting; their inputs and any implementation changes are reported explicitly. Our method additionally receives the target synopsis.

c) Target adaptation.: For Configuration C, target-only and fine-tuned comparisons use the same designated 80% target training subset. Where compatible with a baseline, the fine-grained subgraphs corresponding to those supernodes form its target training data. The remaining 20% are excluded from model optimisation. Local reconstruction is evaluated on the held-out instances, and assembled graphs are evaluated against the full target graph.

For the reported LUXEMBOURG→OLDENBURG comparison, fine-tuning lowers the mean degree discrepancy, GCC relative error, and APL relative error. The change in degree discrepancy is more pronounced than the changes in the two other errors. These results motivate evaluating global structure separately from degree fidelity.

F. Efficiency and Scalability

a) Objective and design.: We assess how runtime and memory change with graph size and identify the dominant generation stage. Table VIII compares inference cost, while Table IX reports the workload breakdown.

TABLE VIII
COMPUTATIONAL COST OF FULL-GRAPH SYNTHESIS. INFERENCE TIME REFERS TO GENERATION OF ONE COMPLETE GRAPH. OOM DENOTES AN OBSERVED OUT-OF-MEMORY FAILURE; DASHES INDICATE UNREPORTED RESULTS.

Dataset	Method	Inference (s) ↓	Peak GPU (GB) ↓
OLDENBURG	Ours	784.7 ± 1.0	19.91*
	DiGress	–	–
	HiGen	–	–
	HiGGs	–	–
	ILE	–	–
OGBL-COLLAB	Ours	–	–
	DiGress	–	–
	HiGen	–	–
	HiGGs	–	–
	ILE	–	–
OBGN-ARXIV	Ours	–	–
	DiGress	–	–
	HiGen	–	–
	HiGGs	–	–
	ILE	–	–

*Maximum observed GPU allocation across component models; not an inference-only peak measurement. Inference-only profiling is pending.

TABLE VI

CROSS-GRAPH TRANSFER AND TARGET ADAPTATION. TARGET-ONLY AND FINE-TUNED MODELS USE THE DESIGNATED 80% TARGET TRAINING SPLIT; LOCAL EVALUATION USES THE HELD-OUT 20%. ZS DENOTES SOURCE-ONLY TRANSFER AND FT DENOTES SOURCE PRE-TRAINING FOLLOWED BY TARGET FINE-TUNING. LOWER DISCREPANCIES ARE BETTER.

$G_s \to G_t$	Method	G_s pretrain	G_t train	Degree W1 ↓	GCC RE ↓	APL RE ↓	Clustering ↓	Orbit ↓
LUXEMBOURG→OLDENBURG	Ours-ZS (Config B)	✓	0%	0.1020	19.39%	23.55%	–	–
	ILE-ZS	✓	0%	–	–	–	–	–
	Ours-Target	–	80%	–	–	–	–	–
	ILE-Target (original ILE)	–	80%	–	–	–	–	–
	Ours-FT (Config C)	✓	80%	0.0248	17.50%	22.57%	–	–
	ILE-FT	✓	80%	–	–	–	–	–
	HiGGs	–	80%	–	–	–	–	–
	HiGen	–	80%	–	–	–	–	–
DBLP→OGBL-COLLAB	Ours-ZS	✓	0%	–	–	–	–	–
	ILE-ZS	✓	0%	–	–	–	–	–
	Ours-Target	–	80%	–	–	–	–	–
	ILE-Target (original ILE)	–	80%	–	–	–	–	–
	Ours-FT	✓	80%	–	–	–	–	–
	ILE-FT	✓	80%	–	–	–	–	–
	HiGGs	–	80%	–	–	–	–	–
	HiGen	–	80%	–	–	–	–	–
CITATION-NETWORK-V19→OBGN-ARXIV	Ours-ZS	✓	0%	–	–	–	–	–
	ILE-ZS	✓	0%	–	–	–	–	–
	Ours-Target	–	80%	–	–	–	–	–
	ILE-Target (original ILE)	–	80%	–	–	–	–	–
	Ours-FT	✓	80%	–	–	–	–	–
	ILE-FT	✓	80%	–	–	–	–	–
	HiGGs	–	80%	–	–	–	–	–
	HiGen	–	80%	–	–	–	–	–

TABLE IX

RECONSTRUCTION WORKLOAD AND GENERATION-TIME BREAKDOWN. $n_{\max}$ IS THE MAXIMUM SUPERNODE SIZE. ADJACENCY WORKLOADS ARE NORMALISED BY n^2.

Dataset	$n_{\max}$	C_{intra}/n^2	C_{inter}/n^2	T_{intra} (s)	T_{inter} (s)
OLDENBURG	79	0.00478	0.01015	149.3 ± 0.9	635.4 ± 0.3
OGBL-COLLAB	–	–	–	–	–
OBGN-ARXIV	–	–	–	–	–

On OLDENBURG, inter-supernode generation takes longer than intra-supernode generation. The small normalised adjacency workloads indicate that reconstruction is decomposed into smaller tasks, but do not establish a corresponding speedup over full-graph diffusion. Runtime also depends on repeated context processing, model architecture, diffusion steps, batching, and implementation overhead.

TABLE VII
END-TO-END GRAPH SYNTHESIS DISCREPANCIES FROM THE REAL TARGET GRAPH. LOWER IS BETTER.

Target	Method	Degree W1 ↓	GCC RE ↓	APL RE ↓	Clustering	Orbit ↓	Spectrum ↓
OLDENBURG	Ours-ZS	–	–	–	–	–	
	Ours-FT	–	–	–	–	–	
	ER	–	–	–	–	–	
	SC-RG	–	–	–	–	–	
	HiGen	–	–	–	–	–	
	HiGGs	–	–	–	–	–	
	ILE	–	–	–	–	–	
OGBL-COLLAB	Ours-ZS	–	–	–	–	–	
	Ours-FT	–	–	–	–	–	
	ER	–	–	–	–	–	
	SC-RG	–	–	–	–	–	
	HiGen	–	–	–	–	–	
	HiGGs	–	–	–	–	–	
	ILE	–	–	–	–	–	
OBGN-ARXIV	Ours-ZS	–	–	–	–	–	
	Ours-FT	–	–	–	–	–	
	ER	–	–	–	–	–	
	SC-RG	–	–	–	–	–	
	HiGen	–	–	–	–	–	
	HiGGs	–	–	–	–	–	
	ILE	–	–	–	–	–	

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